

FINAL REPORT

N100 Financial Intelligence Platform

An ETL Pipeline for Nifty 100 Financial Datasets

Prepared for the Bluestock Fintech Internship

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1. Abstract

This project implements an ETL pipeline that extracts financial datasets from Excel workbooks, validates and normalizes the data, and loads them into a SQLite database.

2. Objectives

The pipeline is designed to extract, transform, validate, and load financial datasets into SQLite while producing audit reports and automated tests. Specifically, the project aims to:

- Extract raw financial data from multiple Excel workbooks
- Transform and normalize the data into a consistent schema
- Validate records for completeness and integrity
- Load the validated data into a centralized SQLite database
- Verify results and generate supporting audit reports

3. Technologies

The following technologies form the core stack used to build, test, and version the pipeline:

Technology	Role in the Pipeline
Python 3.11	Core scripting language for the ETL pipeline
Pandas	Data manipulation, cleaning, and transformation
SQLite3	Lightweight relational database for storage
OpenPyXL	Parsing and reading Excel workbook files
Pytest	Automated unit testing framework
Git & GitHub	Version control and repository hosting

4. Datasets

The pipeline processes twelve structured datasets spanning company profiles, financial statements, and market data:

1. Companies	2. Analysis	3. Balance Sheet
4. Cash Flow	5. Documents	6. Profit & Loss
7. Pros & Cons	8. Financial Ratios	9. Market Cap
10. Peer Groups	11. Sectors	12. Stock Prices

5. Workflow

Data moves through the pipeline in eight sequential stages, from initial ingestion through final testing:

1	Load Excel files
2	Normalize columns
3	Validate data
4	Create database
5	Load tables
6	Verify records
7	Generate audit reports
8	Run Pytest

6. Validation

Before data is committed to the database, each dataset passes through a series of automated validation checks designed to catch structural and quality issues early:

Check	Description
Duplicate row checks	Identifies and flags fully duplicated records within each dataset
Duplicate ID checks	Ensures primary identifiers are unique across all records
Null-value detection	Scans required fields for missing or empty values
Negative-value checks	Flags financial fields containing invalid negative figures
Row-count verification	Confirms loaded row counts match the source datasets

7. Testing

All three Pytest unit tests passed successfully, confirming that the pipeline's core extraction, transformation, and loading logic behaves as expected.

8. Conclusion

The ETL pipeline successfully loads and validates financial datasets into SQLite and is ready for further analysis. The combination of automated validation, testing, and version control provides a reliable and reusable foundation for future extensions of the platform.